

Zhangpen work on long sequence. 8/4/22.

The problem Zhangpen has worked on can be formulated into the following

$$l(w) = E_{\xi_1} f \left(E_{\xi_2} (g(w, \xi_2(w, \xi_1))) \right)$$

$\uparrow$ data $\uparrow$ loss $\uparrow$ probability p is the random variable
 output of g (infer, which is stochastic - $g = p \dots + (1-p) \dots$)

In Zhangpen's work, ξ_2 is a random variable whose distribution also relates to w , but for now let's ignore.

w is the weight to be optimized

ξ_1 : data sample; ξ_2 , random variable in the algorithm

Q1 : How to correctly calculate the 'stochastic'
 ad

$$E_{\xi_1} f \left(E_{\xi_2} [g(w, \xi_2(w, \xi_1)) | \xi_1] \right)$$

$$\xi_1 \text{ given. } \nabla f \dots = E_{\xi_2} \left(\nabla g(w, \xi_2(w, \xi_1) | \xi_1) \right) \nabla f \left(\dots \right)$$

Q1 : if we can accurately compute this, can we do SGD? Yes.

So what we need to compute is

$$E_{\epsilon_2} \left(v_w g(w_1, \epsilon_2(w_2, \epsilon_1)) \right) \cdot \text{of} \left(E_{\epsilon_2} g(\cdot) \right)$$

So the direct way of doing this is to
Sample a lot of times ϵ_2 . $\bar{\epsilon}_2$

Then calculate the average of $g(\cdot)$, denoted as
 $\bar{g}(\cdot)$ (under the current data sample)

Then compute $\text{of}(\bar{g}(\cdot))$

Then sample again.

$$\underline{\underline{\text{Compute } \frac{1}{N} \sum_{(i)} v_w g(w_1, \epsilon_2^{(i)}(w_2, \epsilon_1)) \cdot \text{of}(\bar{g}(\cdot))}}}$$

In practice, not sure how many samples are needed.
could be expensive.